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STRUCTURED CABLING PLAN FOR STUDYLAB

A Structured Cabling Plan Documentation

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CHAPTER 1
INTRODUCTION

Background of the Study

StudyLab is a coworking and study space that promotes productivity and collaboration among its users. With high-speed Wi-Fi and a variety of learning environments to provide users based on their preference and needs. Customers have the option to work outside with free access to fresh air and sunlight on a patio or outdoor workspace or inside the building which is an air-conditioned and well-lit area and has easily accessible power outlets creating an ideal quality of work and study. A private workspace is available for customers seeking peace and quiet which can also be utilized for meetings and small group discussions. A dedicated research room provides users without personal devices access to computers for research purposes. There are areas where only staff have access to, such as a control room, which serves as the central monitoring point for staff, allowing them to maintain a secure environment and also functions as a staff break room. A kitchen and stockroom for preparing meals and storage of ingredients. This diverse spatial configuration provides users with a high degree of flexibility and caters to various learning styles and work preferences. StudyLab provides a unique environment for students, remote workers, and freelancers. To ensure that the environment inside and outside of StudyLab functions smoothly and meets the needs of the users, structured cabling will be developed to have a sturdy and efficient network infrastructure.

An systematic method for creating, implementing, and overseeing a cabling plan is referred to as "structured cabling," which is renowned for its connectivity and scalability (Logeshwaran et al. (2022). A structured method for creating, implementing, and overseeing a cabling strategy is known as structured cabling. It's crucial to comprehend the idea and execution of a structured cabling strategy before creating one. Structured cabling is well-known for supporting scalability and being connected. Therefore, the foundation for a good cabling design is laid by taking the time to comprehend its relevance.

This study sought to develop a structured cabling plan for the company, StudyLab, a multi-functional learning environment designed to promote productivity, collaboration, and well-being for its users. The goal of this study is to develop a structured cabling plan to support StudyLab’s functionality.

Statement of the Problem

The main objective of the study is to develop a structured cabling plan to support StudyLab's functionality to have a robust and efficient network infrastructure. In order to visualize the layout and implementation of the structured cabling for StudyLab, the researchers aim the following objectives:

1. To construct the following floor plans and compute for the cost analysis:
    a. Floor plan
    b. Electrical plan
    c. Cabling plan
    d. Lighting plan
    e. Furniture and fixtures
2. Create a VLSM addressing scheme
3. Simulation using Packet Tracer

Purpose of the Study

In this study, the researchers are tasked to plan and carry out a comprehensive structured cabling plan with the process of making a floor plan, electrical plan, lighting plan, structured cabling plan, and furniture and fixture plan for 3 idealized different buildings of StudyLab. The purpose of this study is to understand the concept and implementation of structured cabling plans.

With the implementation of structured cabling, StudyLab's network performance would improve and simplify the management processes. It also allows an organized network infrastructure that serves multiple purposes. With structured cabling, it enables uncomplicated troubleshooting and maintenance and facilitates efficient data transmission which leads to a smooth operation.

The purpose of this study is to develop a structured cabling plan for 3 idealized buildings of StudyLab. The placement of these 3 buildings would each be placed anywhere from Luzon, Visayas, and Mindanao, specifically in Manila, Cebu, and Davao City. This study will be conducted in Cisco Packet Tracer in the timeframe of May 5 - 6, 2024, which is after the making of the floor plans.

The study will be carried out in a systematic way which involves some important steps. This includes researching and modifying floor plans and identifying the requirements of the institutions or businesses to be housed in buildings, designing and implementing infrastructure plans using advanced technology tools as well as rationalizing budget allocations to ensure cost-effectiveness and functionality.

Scope and Limitations

The study will focus on developing and documenting structured cabling plans for 3 buildings of StudyLab each located in Luzon, Visayas, and Mindanao. The study intended to create floor plans for the buildings, customized cabling plans, and creating reports with final structured cabling plan and cost analysis. However, this study will focus on planning and documentation, not be implemented in real life but serve as representation. The implementation of the structured cabling would be implemented using technological tools or software. Through this process, the researchers will be able to structure a topology and successful cabling plan for the 3 buildings. This study delves into the cabling plans, designed to be adaptable to various functionalities but might require more customization for optimal use.

Definition of Terms

To provide a clearer and better understanding of the context of the study, these are the following terms that are related to the study with the definition that correlates to it:

Structured cabling

A building cabling infrastructure that consists of a number of subsystems.

VLSM

Variable Length Subnet Mask. A means of allocating IP addresses into subnets with different subnet masks.

Infrastructure

The basic physical and organizational structures of a business, region or nation.

Packet Tracer

A visual simulation tool that allows users to create network topologies and imitate modern computer networks.

Topology

A diagram of the arrangement of the physical and logical elements of a network.

CHAPTER 2
REVIEW OF RELATED LITERATURE

Structured Cabling

According to Logeshwaran et al. (2022), Integrated Structured Cabling Systems (ISCS) are indispensable in today's dynamic business environments, serving as the backbone of seamless communication. They assert that ISCS goes beyond mere cable connections, emphasizing the creation of a robust network capable of keeping communication flowing smoothly. With ISCS in place, organizations can ensure swift and uninterrupted communication among computers, phones, and other devices. By adhering to international standards and leveraging cutting-edge technologies such as fiber optics, ISCS ensures that business communication stays reliable even as technology changes over time.

Electromagnetic interference

In their investigation, Akshiti (2019) explored the impact of wire separation on electromagnetic interference (EMI) within structured cabling systems. Their study aimed to understand how varying the distance between wires influences the occurrence and severity of EMI, which is critical for ensuring the reliability of data transmission.

Electromagnetic interference occurs when electricity flowing through one wire generates unwanted signals in nearby wires, potentially disrupting signals and corrupting data. As explained by Faraday and Maxwell (1855), this phenomenon, known as electromagnetic induction, occurs due to changing magnetic fields inducing electric currents in adjacent wires. In structured cabling systems where cables are densely packed, EMI poses a significant challenge. One of the many things that radiate electromagnetic waves are fluorescent lights, contributing to network interference.

A study conducted by Merenciano et al. (2022) investigated whether different types of light sources affect Wi-Fi signals in the 2.4 GHz band. The tests were conducted in a laboratory at a university in Brazil with a controlled environment and seven kinds of light sources. A tool called Iperf3 was utilized to perform the tests. A total of three tests were performed:

- **Spectrum analysis:** With the use of wi-spy tool, this test identified how each light affects Wi-Fi channels.

- **Data transfer rate:** Using two routers, this test identified whether the lights slowed down the Wi-Fi.
- **Multiple lights test:** The tests were done again, but this time with multiple lights, to better identify changes in the rate of data transmission.

The results showed that, despite some external factors added, in this case different types of lights, there was no interference discovered caused by the tested lights found in the 2.4 GHz band. Additionally, even though the data transfer rate fluctuated slightly at some test points, it was concluded that it was not caused by the lights. The study concluded that common lighting technologies caused no significant interference with 2.4 GHz Wi-Fi signals, however, the impact on 5 GHz Wi-Fi and other common bands was not investigated. This may be the reason why no significant interference from fluorescent lights was discovered during the tests, as most fluorescent lights only emit radio waves in the MHz band range. Although very unlikely, MHz frequency bands may still cause interference to WIFI networks (Jose Montero Martin, 2024). This is why the use of LED lights was preferred for this project, as they do not emit radio waves, are more energy efficient, and are safer as they do not contain mercury, which could be toxic to humans.

Recognizing this, the researchers took a proactive approach, including wire separation, to address this issue. This includes separation of wires in order to mitigate Electromagnetic interference, thereby safeguarding the integrity of data transmission. Be more detailed daw diri

Internet of Things (IoT)

The Internet of Things (IoT) involves connecting everyday objects to the internet, enabling them to collect and exchange data autonomously. This interconnected web of devices holds immense promise, offering unparalleled opportunities for efficiency improvements, cost reductions, and innovation across various industries.

Meola (2016) emphasizes the broad adaptation of IoT solutions across sectors, with business leading the way due to the multifaceted benefits, including lowered operating costs, increased productivity, and market expansion opportunities. This sentiment is supported by Rouse and DelVecchio (2015), who also highlighted the transformative impact of IoT in the healthcare industry through applications like remove patient monitoring and real-time location services, promising cost reduction and increased patient care.

However, alongside the opportunities, challenges and risks accompany IoT adoption. Mukherjee (2016) warns of cybersecurity threats associated with IoT, emphasizing the importance of robust security measures. Moreover, Dobbs, Manyika, and Woetzel (2015) caution against over-hyped expectations and underline the need for strategic planning and risk management to fully leverage IoT's potential.

Ergonomic Design of Furniture for Comfort and People with Musculoskeletal Disorders

A study by Afrifah & Glalah in 2021 inspected university classroom furniture to evaluate their comfortability and took into account whether optimizations could be made, especially for those with musculoskeletal disorders. They recommended that classroom furniture, such as chairs and tables, and their dimensions should be based on student anthropometry, which refers to the measurements of students' bodies, including height, leg length, torso size, and the like. Chair seat heights and depths should be designed to be adjustable in order to accommodate for a range of student leg and torso lengths. Continuing about chairs, backrests should be an adequate height to provide the right support for students' shoulders. For tables, their height must be set at a comfortable level, preferably at a level that allows students to rest their elbows when doing tasks. To implement a more conducive learning environment, these criteria for comfort will be evaluated when choosing the furniture and fixtures for StudyLab.

CHAPTER 3
TECHNICAL REFERENCES

This technical documentation is intended for network engineers and network planners, which provides the purpose of the design, implementation, and management of structured cabling systems. This provides the best practices, as well as to guide the planning and implementation of structured cabling projects.

Lucidchart

Lucidchart is a web based diagram application that enables users to visually collaborate on drawing, revising, and sharing diagrams in order to improve processes, systems, and organizational structures. Users can create floor plans for various sized buildings as Lucidchart allows for scaling objects. With the premade shapes provided, users can also create custom shapes to accurately represent specific features of the buildings. The researchers are already familiar with using Lucidchart and being proficient in operating the software in order to save time and effort in creating floor plans compared to using other software and manual floor plan creation.

Cisco Packet Tracer

Cisco Packet Tracer is a product of Cisco Systems that has been widely used by schools around the world, especially to the computer networking students. It is a tool for teaching and learning of computer network courses by providing visualizations and simulations allowing the students to practice and experiment with network tasks. It provides the experiences and capability to make own network topologies in a visual way by connecting and clustering network devices. By using Cisco Packet Tracer, it allows us to visualize the network infrastructure we're planning for each building. We can use the available elements or devices such as routers, switches, PC's and others, to represent the core network components we'll be incorporating into our structured cabling plan, providing a visual representation of how data will flow through the network.

CHAPTER 4
METHODOLOGY

Floor Plan

Considering that this building is a Co-working Space, it is crucial for the floor plan to be designed to be spacious and organized simply to ensure that the environment is conducive to learning and there are minimal distractions. Additionally, there is a designated control room that will contain the routers and switches for efficient implementation of structured cabling.

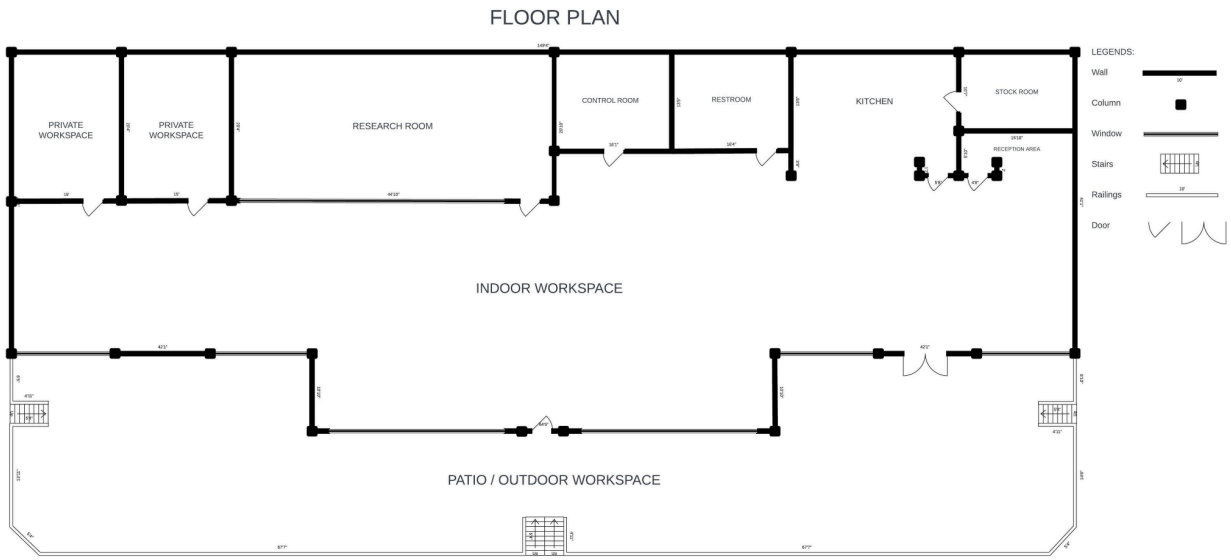

Figure 1. StudyLab Floor Plan

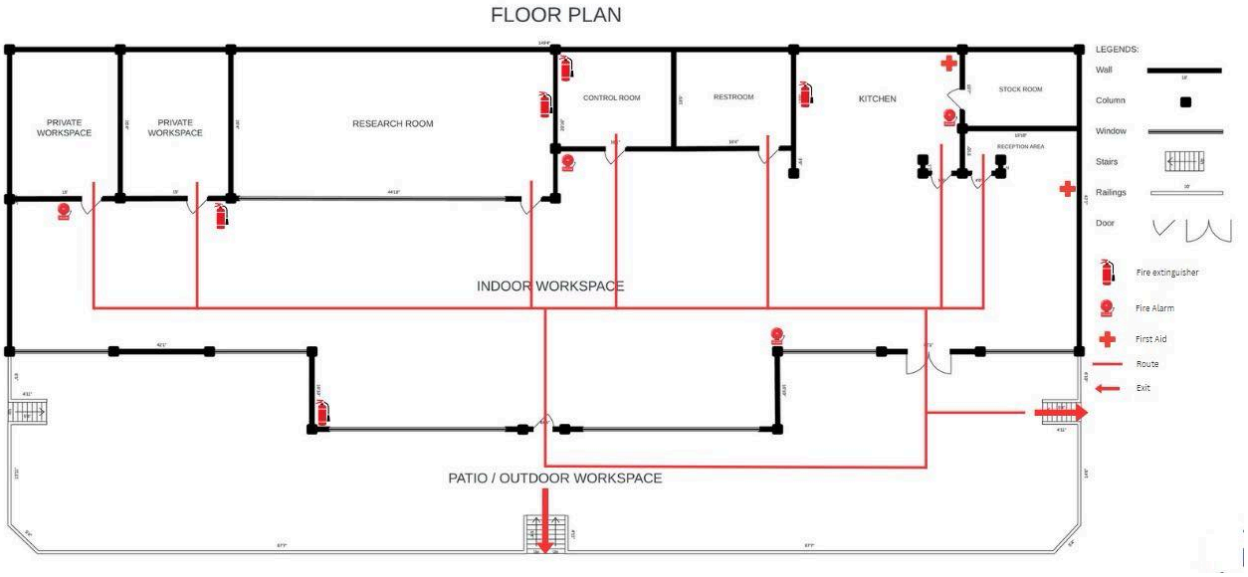

Figure 2. StudyLab Evacuation Map

The importance of cost analysis for this floor plan is just as important as the designing phase of this floor plan. This sets expectations on how much budget should be prepared in order to implement this floor plan. Below is the analysis of the cost of the goods required for this plan.

Table 1

Floor Plan Cost Analysis in Php

Floor Plan Cost Analysis					
Item	**Supplier**	**Type**	**Quantity**	**Cost (Php)**	**Total**
Ready Mix Concrete	Suprea Phils. Development Corp.	4000 PSI	60	5,750.00	345,000
Hollow Blocks	Top-Most Hardware	6”	5,510	25.00	137,750
Steel Reinforcement Bar	Cebu Steel Corporation	Deformed Steel Bars	3,100	40.00	124,000
Wall paint	Boysen Philippines	Quick Drying	105	799.50	83,947.50
Floor Tiles	CitiHardware	Tiles Espana Premium CC 60X60F	2,780	174.00	483,720
Wood	Top-Most Hardware	Wood Planks	175	1,071.00	187,425
Doors	Top-Most Hardware	Solid Wooden Doors Atlanta	11	5,875.00	64,625
Windows	Carousell	Clear Tempered Glass	12	2,500.00	30,000
Roof	Cebu Home and Builders	ADCO Ceiling Panel	138	330.00	45,540
				TOTAL:	1,502,007.50

Electrical Plan

The electrical wiring for this establishment is designed to be safe and have minimal electromagnetic interference to lessen possibilities of slowing down and losing the network signals. This can be ensured by embedding the electrical wires on the ground, separate from the cables for the networks.

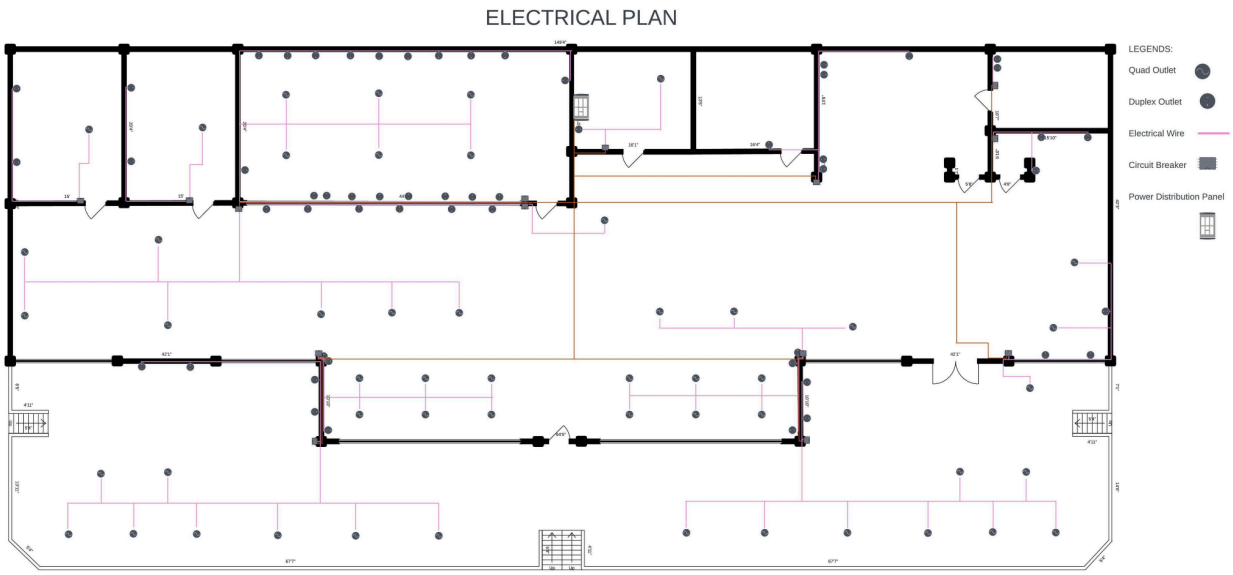

Figure 3. StudyLab Electrical Plan

Below is the analysis of the cost of the goods mentioned above with the required specifications recommended for this electrical plan.

Table 2

Electrical Plan Cost Analysis in Php

Electrical Plan Cost Analysis					
Item	**Supplier**	**Type**	**Quantity**	**Cost (Php)**	**Total**
Quad outlet	Felco Store	ROYU 4 Gang Surface Type	51	90.00	4,590.00
Duplex outlet	Wilcon Depot	Alphalux 2 gang Duplex Universal Outlet	54	139.75	7,546.50
Electrical wire	V.SY Electronics	TKK Cable Duplex Flexible	1000m	55 per 5 meters	11,000.00
Circuit breaker	Wilcon Depot	Schneider Ez9f56220 Easy9	14	365.00	5,110.00
Power Distribution Panel	Giant Electric	Zhegui Electrical Low Voltage Metal Wall Mounted	1	3,430.00	3,430.00
				TOTAL:	31,676.50

Cabling Plan

The cabling plan for this establishment is the most important aspect in ensuring the network to be effective and efficient in terms of data transmission rate and minimal possibilities for packet losses. This was also designed to have minimal electromagnetic interference especially from the electrical and lighting wires, which was done by separating these wirings from the other wires. Cabling wires are to be embedded on the wall while electrical wires are embedded below the ground. Some of the most important devices and materials used to develop this network are Cisco routers and switches, serial, cross-over, and straight-through cables.

Furthermore, to ensure an efficient network connection and competent internet connectivity, this was designed to be a structured cabling plan wherein there are subnetworks. For this network, there are two main subnetworks, one for the administrators, and one for the public hosts. These two subnetworks are connected by a router, which will be stored in the control room.

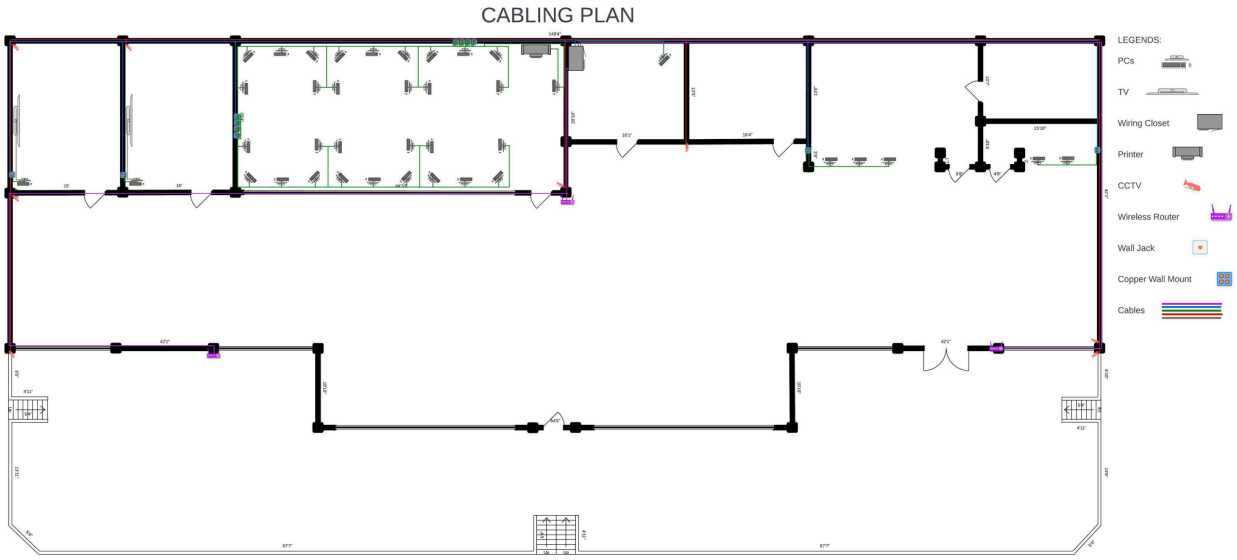

Figure 4. StudyLab Cabling Plan

Below is the analysis of the cost of the goods required to develop this cabling plan. Despite being costly, this establishment chooses to use CISCO products to ensure quality performance, reliability, and have access to advanced features that are beneficial to the current network, while also reliable enough to be able to expand the network, as these products are also known for their scalability.

Table 3

Cabling Plan Cost Analysis in Php

Cost Analysis					
Item	**Supplier**	**Type**	**Quantity**	**Cost (Php)**	**Total**
NVR	DigitalCCTV	DS-7604 NI-E1 series Standalone NVR	1	8,500.00	8,500.00
Switch	CISCO	CISCO Catalyst 2960X-24TS-LL	3	55,686.71	167,060.13
Router	CISCO	CISCO 2911/K9	1	78,421.70	78,421.70
Wireless Router	CISCO	TP-Link Archer AX21	3	4,280.00	12,840.00
Computers	HP	EliteOne 840 G9	39	55,451.99	2,162,589.00
TV	Samsung	43" Full HD T5202 Smart TV	2	20,399.00	40,798.00
Wiring Closet	Benson Computer	Hard Rack Data Cabinet	1	39,990.00	39,990.00
Printer	CompLink	Epson L18050	1	28,925.00	28,925.00
CCTV	iPrice	Hikvision DS-2CE16D0T-IRPF CCTV	8	790.00	6,320.00

Wall Jack	Belmont	RJ 45 wall jack	39	175.00	6,825.00
Copper wall mount	InfoBahn	Premium Line Wall Mount Cabinet Width 600mm	8	6,730.00	53,840.00
Straight-through Cables	Belkin	CAT6 Ethernet Patch cable	200m	39.67 per meter	7,934.00
Cross-over Cables	Belkin	CAT6 Gigabit Ethernet	100m	196.50 per meter	19,650.00
Server	CompLink	Acer TC-1770 DT.BK7SP.004	1	66,999.00	66,999.00
				TOTAL:	2,700,691.83

Lighting Plan

This shows how the establishment will be illuminated. As fluorescent lights emit radio waves which can interfere with the network connection, this establishment will utilize LED lights as these do not emit electromagnetic waves which could potentially cause network connectivity issues and loss of network signals. Moreover, these wiring will be embedded in the ceiling for convenience and avoid EMI to the network.

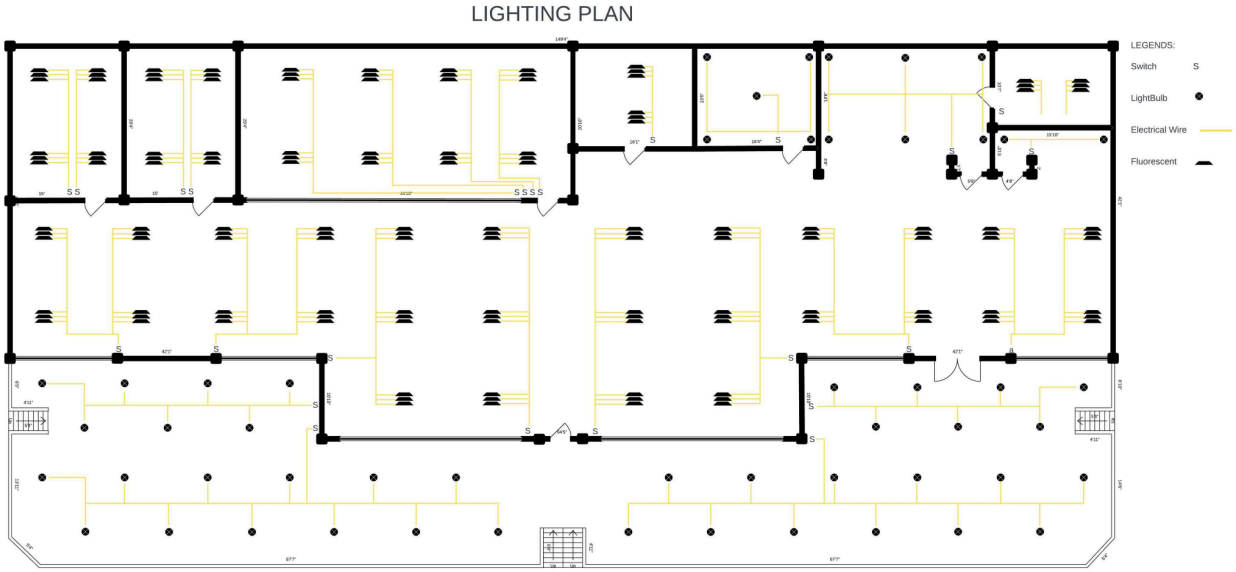

Figure 5. StudyLab Lighting Plan

Below is the cost analysis for the establishment's lighting plan, showing the required materials and their costs.

Table 4

Lighting Plan Cost Analysis in Php

Cost Analysis					
Item	**Supplier**	**Type**	**Quantity**	**Cost (Php)**	**Total**
LED light bulb	Ace hardware	Feit A19 E26	51	16.99	866.50
Fluorescent Tube Light	Ace hardware	Feit Linear T8/T12	154	44.99	6,928.50
Electrical Wire	RS Philippines	Two core wire	300 meters	64.25 per meter	19,275
Switch	Ace hardware	Leviton Decora	24	49.99	1,199.76
				TOTAL:	28,269.76

Furniture and Fixtures

At StudyLab, the furniture choices are carefully made to prioritize client comfort and concentration. Drawing from Afrifah & Glalah's 2021 study on ergonomic design, we focus on adjustable chair heights and comfortable table levels to support optimal posture, as well as minimalism in design to promote productivity and minimize distractions. This commitment ensures a conducive learning environment where clients can thrive.

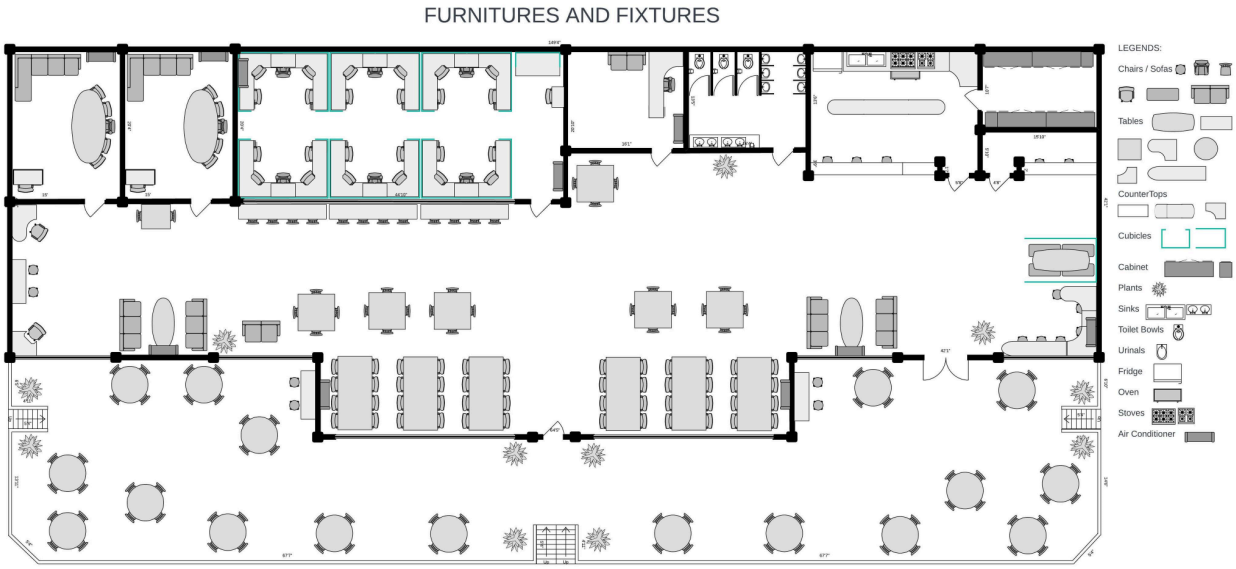

Figure 6. StudyLab Furniture and Fixtures

Presented below is the list of furniture and fixtures to be purchased for StudyLab and along with their corresponding prices.

Table 5

Furniture and Fixtures Cost Analysis in Php

Cost Analysis					
Item	**Supplier**	**Type**	**Quantity**	**Cost (Php)**	**Total**
Love Seat	Mandaue Foam	Eric Sofa	2	8,330.00	16,660
3 Seat Sofa	Mandaue Foam	Camrose Sofa	4	14,140.00	56,560
L-shaped couch	Mandaue Foam	Eric Sofa	2	14,490.00	28,980
		Marlowe			

L-shaped Desk	Mandaue Foam	L-shape Desk	16	2,905.00	46,480
Writing Desk	Mandaue Foam	Archie 90cm Desk	9	3,850.00	34,650
Square Table	Mandaue Foam	Irvine 75cm	7	4,585.00	32,096
Long Table	Mandaue Foam	Elland 8 seater	10	10,745.00	107,450
Round Table	Mandaue Foam	Brooke 150cm	17	10,990.00	186,830
High Table	Mandaue Foam	Gerik Bar Table	3	2,835.00	8,505
Coffee Table	Mandaue Foam	Youset Coffee Table	2	2,646.00	5,292
Regular chair	Mandaue Foam	Skylar chair	104	1,260.00	131,040
Barstool	Mandaue Foam	Asher Plastic Bar chair	17	2,065.00	35,105
Office chair	Mandaue Foam	Gregor Low Back	94	4,235.00	398,090
Counter tops	Wilcon Depot	P.techNatural granite	3	6,890.00	20,670
Cabinet	Mandaue Foam	Cabello Base Cabinet	6	6,650.00	39,900
House Plant	Ikea	Zamioculcas	11	890.00	9,790
Air-conditioner	Wilcon Depot	Eakon Series	10	25,330.00	253,300
Stoves	Wilcon Depot	2 Burner Glass Stove	2	3,090.00	6,180
Oven	Wilcon Depot	Built-In oven	1	18,506.00	18,506
Fridge	Wilcon Depot	Fujidenzo Rdd-35	1	8,998.00	8,998
Sink	Wilcon Depot	Single Bowl Sink	3	6,440.00	19,320
		Pozzi Top			

Urinal	Wilcon Depot	Urinal Wall	6	3,220.00	19,320
Toilet Bowls	Wilcon Depot	Two piece water closet	3	7,950.00	23,850
				TOTAL:	1,507,572

Table 6

Overall Budget Plan

Overall Total of Cost Analysis in Php	
Floor Plan Cost Analysis	1,502,007.50
Electrical Plan Cost Analysis	31,676.50
Cabling Plan Cost Analysis	2,700,691.83
Lighting Plan Cost Analysis	28,269.76
Furniture and Fixtures Cost Analysis	1,507,572
TOTAL:	**5,770,217.59**

Variable Length Subnet Mask Addressing Scheme

The VLSM addressing scheme allows subnetworks to have their own network addresses and subnet masks, and lessens the number of wasted IP addresses. Hence, the establishment will be implementing this technique in assigning IP addresses. The following VLSM addressing scheme will be applied:

Device	Interface	IPv4 Network Address	IPv6 Network Address
Router 0	int g0/0	192.1.168.0/26	2001:db8:acad:1::0/64
Router 0	int g0/1	192.1.168.64/58	2001:db8:acad:2::0/64

Static Routing

For the IPv6, static was used to allow the routers to communicate. Static routing is a manually configured routing entry

Protocols Used

Routing Protocols

Routing Protocols for IPv4 to be used are RIPv2 and default static route will be used for the cabling plan. RIP will be implemented within the networks of the buildings while Edge Router will use default static routing to the other buildings and the ISP. Routing Protocols for IPv6 will only use default static routing.

Dynamic Host Configuration Protocol(DHCP)

The main routers of each building will serve as DHCP routers for IPv4. For IPv6, SLAAC will be implemented to the different networks of each building.

File Transfer Protocol(FTP)

The servers of each building will have FTP services to enable file transfers from one building to another.

CHAPTER 5
CONCLUSIONS AND RECOMMENDATIONS

Summary

The study focused on developing a structured cabling plan for StudyLab, a co-working space that promoted productivity, collaboration and focus by providing a learning conducive environment. This encompasses the different stages of development for the establishment, including a floor plan, an electrical plan, cabling plan, lighting plan, and a furniture and fixtures plan. This also covers the cost analysis of each plan, including the total budget plan for this project. The structured cabling plan for this project aims to support and enhance the experience of clients, as well as providing an effective and efficient network connection.

After performing a review of related literature, these are some of the key findings that are ought to be essential in building this project:

- There is electromagnetic interference produced by some electrical devices and wirings, thus it is important to separate the electrical and lighting wirings from the cabling wiring to prevent causing network interference.
- It is important to consider the health and comfort of the clients to ensure a safe and learning conducive environment.

Moreover, this study used the following tools and techniques to develop the plans and simulation of the project:

- Lucidchart for creating the floor plans,
- Cisco Packet Tracer for simulation of the network,
- Variable Length Subnet Mask (VLSM) for computing the network addresses, and
- DHCP to automatically allocate IP addresses for enhanced network management.

Overall, the study aims to provide a learning conducive environment for people who want to work and study productively, through a well-designed structured cabling plan.

Conclusions

To conclude, the total sum of the overall cost analysis is ₱5,770,217.59 with the cabling plan as the cost which is ₱2,700,691.83. All of this accumulates to the construction cost, setup of the whole project and the materials needed. However with the cost being so high, the labor, installation and maintenance cost is still not being accounted for. The initial approach was to only base hypothetically, but we recognize once it is implemented in reality, modifications are required to fit in with the constraints of real-world implementation.

Recommendations

In accordance with the findings and conclusions, this study found that in developing a similar project, the following are recommended:

- **In-depth Cost Analysis.** This study only accounted for the materials and devices required for each component. It is advisable to perform an in-depth analysis that would include labor, installation, and maintenance costs. This is necessary to have a better understanding of the overall budget plan for the project.
- **In-depth Simulation and Testing.** This study was only simulated using Cisco Packet Tracer, hence needing a more in-depth study that would account for different conditions and environment situations.
- **Risk Assessment.** It is also necessary to identify potential risks in various aspects, including security, calamities, and other factors, in order to be able to address these issues.

Applying these recommendations for future similar projects may help developers in their decision-making process in creating a structured cabling plan that encompasses floor plans, electrical plans, cabling plans, lighting plans, and furniture and fixtures plans that would be beneficial in achieving their own goals.

References

Aleena, A. (2019). Mitigating Electromagnetic Interference (EMI): Strategies for Reducing Interference Between Power and Communication Lines. Scribd. https://www.scribd.com/document/405734430/11-chapter-2-pdf

Bécherrawy, T. (2013). Electromagnetism: Maxwell equations, wave propagation and emission. John Wiley & Sons. https://books.google.com/books?hl=en&lr=&id=Z80DySvAVFkC&oi=fnd&pg=PA9&dq=+Faraday+and+Maxwell++electromagnetic+interference&ots=Jxk4VE5b5H&sig=_Cwt_mxmk8v0csDI09Ampo1mRtY

Dobbs, R., Manyika, J., & Woetzel, J. (2015). The Internet of Things: Mapping the value beyond the hype. McKinsey & Company

Jose Montero Martin. (2024, March 18). Does lighting affect the performance of your WiFi? Linkedin.com.https://www.linkedin.com/pulse/does-lighting-affect-performance-your-wifi-jose-montero-martin-oijkf

Logeshwaran, J., Ramkumar, M., Kiruthiga, T., & Ravi, S. P. (2022). The role of integrated structured cabling system (ISCS) for reliable bandwidth optimization in high-speed communication network. *ICTACT Journal on Communication Technology,* 13(1), doi: 10.21917/ijct.2022.0389

Mukherjee, A. (2016). How IoT Might Cause Cyber Security Threats in 2017. DZone. Rouse, M. and DelVecchio, A, (2015). IoMT (Internet of Medical Things) or healthcare IoT." TechTarget

Noor, N. M. M., Yayao, N., & Sulaiman, S. (2018). Effectiveness of using cisco packet tracer as a learning tool: A case study of routing protocol. International Journal of Information and Education Technology, 8(1), 11–16. https://doi.org/10.18178/ijiet.2018.8.1.1004

Silva, Luana & Oliveira, Lorenna & araujo, julia & Silva, Joacir & Fortes, Marcio. (2023). A study on the impact of EMI lighting systems on 2.4 GHz Wi-Fi communication networks. Ingeniare. Revista chilena de ingeniería. 30. 693-704. 10.4067/S0718-33052022000400693.